

MAJOR TECHNICAL PROJECT

Study of Infrared Signature from Aircraft Plume

Complete Implementation Guide in OpenFOAM

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Final Year B.Tech Project Report

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1.1 Problem Statement

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Infrared (IR) guided Anti-Aircraft Missiles, particularly Man Portable Air Defense Systems (MANPADS), pose a significant threat to military aircraft. The exhaust plume ejected from aircraft engines is one of the primary sources of IR signature, making the aircraft detectable by adversary sensors. This project aims to develop a computational feature within the OpenFOAM framework to estimate the IR radiation emitted by the exhaust plume of a military aircraft.

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1.2 Objective

18

The primary objective is to build a standalone IR signature estimation solver in OpenFOAM that can compute the spectral radiation intensity arriving at a ground-based sensor from an aircraft exhaust plume. The solver must handle non-gray (wavelength-dependent) radiation behavior and utilize spectroscopic data from the HITEMP database for participating gaseous species (CO_2 , H_2O , CO).

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1.3 Scope of Work

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2. Theoretical Background

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2.1 Radiative Transfer Equation (RTE)

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The radiative transfer equation for a participating medium describes how spectral intensity changes along a ray path due to absorption, emission,

and scattering. The general monochromatic RTE is:

20

$$dI_\eta/ds = \kappa_\eta \cdot I_{b\eta} - \beta_\eta \cdot I_\eta + (\sigma_\eta/4\pi) \int I_\eta(\hat{s}_i) \Phi_\eta(\hat{s}_i, \hat{s}) d\Omega_i$$

20

Where I_η is the spectral intensity along direction $\hat{s}$, $I_{b\eta}$ is the Planck blackbody function, κ_η is the spectral absorption coefficient, σ_η is the spectral scattering coefficient, β_η is the extinction coefficient ($\kappa_\eta + \sigma_\eta$), and Φ_η is the scattering phase function.

20

For gaseous media in aircraft plumes, scattering is negligible compared to absorption and emission. The simplified RTE becomes:

20

$$dI_\eta/ds = \kappa_\eta \cdot I_{b\eta} - \kappa_\eta \cdot I_\eta$$

20

This equation states that intensity increases by emission ($\kappa_\eta \cdot I_{b\eta}$) and decreases by absorption ($\kappa_\eta \cdot I_\eta$) along the path.

20

2.2 Gray vs. Non-Gray Radiation

20

Gray Radiation: Assumes the absorption coefficient κ is constant across all wavelengths. The emitted radiation follows a scaled version of the Stefan-Boltzmann law. This simplification allows a single RTE solve but is inaccurate for gaseous media where absorption is strongly wavelength-dependent.

20

Non-Gray Radiation: Absorption and emission properties vary with wavelength. Gaseous species like CO_2 and H_2O absorb and emit radiation in specific spectral bands, not uniformly across all wavelengths. Accurate IR signature estimation requires solving the RTE separately for each spectral band of interest, using wavelength-dependent absorption coefficients from spectroscopic databases.

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2.3 Important IR Bands

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2.4 HITEMP Database

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The High-Temperature Molecular Spectroscopic Database (HITEMP) provides line-by-line spectroscopic data for molecular species at high temperatures (accurate up to 4000 K and 80 bar). Each spectral line transition is stored in 160-character ASCII format containing parameters such as molecule number, vacuum wavenumber, line intensity, air-broadened halfwidth, self-broadened halfwidth, lower-state energy, and temperature-dependence coefficient. The reference conditions are 296 K and 1 atm. Mathematical models are required to convert these reference parameters to actual thermodynamic conditions encountered in the plume.

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3. Complete Implementation Steps

22

This section provides a detailed, step-by-step guide for implementing the IR signature estimation feature in OpenFOAM, from initial setup to final post-processing.

22

3.1 Phase 1: Environment Setup

22

3.1.1 OpenFOAM Installation

22

Install OpenFOAM v2112 or later. Source the environment:

22

```
source /opt/openfoam/etc/bashrc
```

22

Verify the installation by running a simple tutorial case and confirm that the fvDOM radiation model is available.

22

3.1.2 Project Directory Structure

22

Create the project directory structure following OpenFOAM conventions:

22

```
cd $FOAM_RUN
22
mkdir -p irSignature/{constant/polyMesh,system,0}
22
mkdir -p irSignature/hitemp_data
22
mkdir -p irSignature/scripts
22
```

The irSignature directory will be the main case directory. The hitemp_data folder will hold processed absorption coefficient tables, and scripts will contain Python utilities for HITEMP processing and data mapping.

22

3.1.3 Download HITEMP Data

22

Download the HITEMP line-by-line data files for the target species from <https://hitran.org/hitemp/>. You need the .par files for H₂O, CO₂, and CO at minimum. These are large files (H₂O alone can be several GB). Store them in the hitemp_data directory.

22

3.2 Phase 2: Mesh Generation

23

3.2.1 Initial Plume Mesh

23

Create a cylindrical mesh representing the aircraft exhaust plume. Use blockMesh with a wedge geometry for axisymmetric cases or a full 3D cylinder. The mesh must be fine enough to resolve the temperature and species concentration gradients near the plume core.

23

Key file: system/blockMeshDict

23

Define vertices for a cylinder aligned along the x-axis. Typical dimensions: length 0.3–0.5 m (depending on plume data extent), radius 0.015 m (based on the mole fraction profiles showing where species concentrations drop to near-zero). Use grading to concentrate cells near

the axis where gradients are steepest.

23

blockMesh

23

After running blockMesh, verify the mesh quality:

23

checkMesh

23

Look for no failed checks, reasonable cell volumes, and acceptable non-orthogonality values.

23

3.2.2 Extracting Region of Interest

23

The full plume mesh may contain regions where species concentrations are negligible (contributing zero IR emission). Extract only the active region using topoSet and subsetMesh.

23

Step 1: Examine the mole fraction profiles from the plume CFD data. Identify the axial and radial extent where H₂O and CO₂ mole fractions are above a threshold (e.g., 1% of peak).

23

Step 2: Create system/topoSetDict defining a cylinderToCell source with the identified dimensions.

23

Step 3: Run the extraction:

23

topoSet

23

subsetMesh plumeCore -overwrite

23

This produces a trimmed mesh containing only the radiatively active plume region.

23

3.2.3 Creating the Atmosphere Mesh

23

The atmosphere mesh extends from the plume region boundary down to the ground-based sensor. This domain captures the attenuation of the plume's IR radiation as it propagates through the atmosphere.

23

Method: Create a separate OpenFOAM case with blockMesh defining a conical or cylindrical volume from the plume exit to the sensor location (typically 1–5 km away). The mesh can be coarser than the plume mesh since atmospheric properties vary slowly.

23

Set atmospheric conditions in this region: ambient temperature (~288–300 K depending on altitude), atmospheric pressure (1 atm at ground, decreasing with altitude), and ambient H₂O mole fraction (~1–2% near ground).

24

3.2.4 Merging the Meshes

24

Combine the plume mesh and atmosphere mesh into a single computational domain using OpenFOAM's mesh manipulation utilities:

24

```
mergeMeshes . ../atmosphereCase -overwrite
```

24

```
stitchMesh -overwrite masterPatch slavePatch
```

24

After merging, run checkMesh again to verify the combined mesh is valid. Pay attention to the interface between plume and atmosphere regions — ensure faces are properly connected with no gaps or overlaps.

24

3.3 Phase 3: Data Mapping

25

3.3.1 Mapping Plume Flow-Field Data

25

The plume flow-field data (temperature T , pressure p , and species mole fractions X_{H_2O} , X_{CO_2} , X_{CO}) from a prior CFD simulation or

experimental measurements must be mapped onto the computational mesh.

25

Option A — mapFields: If the source data is from another OpenFOAM case, use the built-in mapFields utility:

25

```
mapFields <sourceCase> -consistent
```

25

Option B — Custom mapping utility: If the source data is in tabular format (CSV, text files), write a custom C++ utility or use codedFixedValue boundary conditions. The utility reads the data file, builds an interpolation table indexed by axial position (x) and radial distance (r), and assigns values to each mesh cell center:

25

```
// For each cell: T[cellI] = interpolate(x, r, tempTable)
```

25

```
// Similarly for p, XH2O, XCO2, XCO
```

25

After mapping, the 0/ directory should contain the following volScalarField files: T, p, XH2O, XCO2, XCO. Verify the mapping by opening the case in ParaView and visually inspecting the fields.

25

3.3.2 Processing HITEMP Data

25

This is one of the most computationally intensive and critical steps. The goal is to convert the line-by-line HITEMP data into band-averaged absorption coefficients that can be used by the RTE solver.

25

Step 1 — Parse the .par files: Read the 160-character fixed-width ASCII format. Extract for each spectral line: wavenumber (ν), line intensity (S) at 296 K, lower-state energy (E''), air-broadened halfwidth (γ_{air}), self-broadened halfwidth (γ_{self}), and temperature-dependence exponent (n).

25

Step 2 — Temperature scaling: Scale the line intensity from the reference temperature (296 K) to the local temperature T using the partition

function ratio and the Boltzmann factor:

25

$$S(T) = S(296) * [Q(296)/Q(T)] * \exp(-c2 * E''/T) / \exp(-c2 * E''/296)$$

25

$$* [1 - \exp(-c2 * v/T)] / [1 - \exp(-c2 * v/296)]$$

25

Step 3 — Line broadening: Calculate the Voigt profile for each line, accounting for pressure broadening (γ_{air} , γ_{self}) and Doppler broadening at the local temperature.

25

Step 4 — Band averaging: Sum contributions from all spectral lines within each band of interest (e.g., 2000–3333 cm^{-1} for the MWIR 3–5 μm band, 1000–2000 cm^{-1} for the strongest fundamental band) to get a band-averaged absorption coefficient.

25

Step 5 — Generate lookup table: Create a lookup table of κ as a function of (T , p , X_{species}). Store this as a structured text file that the OpenFOAM solver can read at runtime.

25

3.3.3 Mapping Absorption Coefficients onto the Mesh

25

Using the lookup table from the previous step and the local T , p , and mole fraction values at each cell center, compute the absorption coefficient κ for each spectral band at every cell. Write these as volScalarField files (e.g., kappaBand0, kappaBand1, etc.) in the 0/ directory. You can use a custom utility or a codedFixedValue approach for this mapping.

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3.4 Phase 4: Modifying the Radiation Solver

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3.4.1 Isolating the fvDOM Solver

27

OpenFOAM's built-in fvDOM (finite volume Discrete Ordinates Method) solves the RTE in gray mode by default. The first step is to copy it into your user directory for modification:

27

```
mkdir -p $WM_PROJECT_USER_DIR/src/myRadiation
```

27

```
cp -r $FOAM_SRC/thermophysicalModels/radiation/
```

27

```
radiationModels/fvDOM/* \
```

27

```
$WM_PROJECT_USER_DIR/src/myRadiation/
```

27

Copy the associated Make/ files and adjust include paths to compile against the OpenFOAM libraries.

27

3.4.2 Adding Multi-Band (Non-Gray) Support

27

The core modification is to loop over multiple spectral bands instead of solving a single gray RTE. In the main solve routine of fvDOM.C, implement the following changes:

27

Band definition: Read band limits (wavenumber ranges) from constant/radiationProperties. Store as a list of pairs: bandLow[i], bandHigh[i].

27

Planck function per band: For each band, compute the Planck function integrated over the band limits at each cell's temperature. This gives the local blackbody emission for that specific spectral band.

27

Band-specific κ : Read or compute the band-averaged absorption coefficient for each band from the mapped fields or lookup table.

27

RTE solve loop: For each band b and each discrete ordinate direction d , solve:

27

$$\text{div}(\mathbf{J}_d, \mathbf{I}_{\{b,d\}}) + \kappa_b * \mathbf{I}_{\{b,d\}} = \kappa_b * \mathbf{I}_{b_band}(T)$$

27

Where $\mathbf{J}_d$ is the direction vector for ordinate d , $I_{\{b,d\}}$ is the intensity for band b in direction d , κ_b is the band-averaged absorption coefficient, and $I_{b_band}(T)$ is the Planck function integrated over band b .

27

Storage: Create `volScalarField` arrays indexed by `[band][ray]` to store the intensity fields for each band and direction.

27

3.4.3 Compilation

27

Create a `Make/files` and `Make/options` in your modified radiation directory. Compile the custom library:

27

```
cd $WM_PROJECT_USER_DIR/src/myRadiation
```

27

```
wmake libso
```

27

This produces a shared library (e.g., `libmyRadiation.so`) that you reference in your case's `system/controlDict` via the `libs` directive.

27

3.5 Phase 5: Case Setup and Simulation

28

3.5.1 constant/radiationProperties

28

Configure the radiation model to use your modified `fvDOM` with band definitions:

28

```
radiation      on;
```

28

```
radiationModel fvDOM;
```

28

```
fvDOMCoeffs
```

28

```
{
```

28

```

    nPhi  4;    // azimuthal angles
28
    nTheta 4;    // polar angles
28
    maxIter 10;
28
    tolerance 1e-4;
28
    }
28
    nBands 3;
28
    bandLimits
28
    (
28
    (2000 3333) // MWIR: 3-5 um
28
    (1000 2000) // Strongest fundamental band
28
    (714 1250) // LWIR: 8-14 um
28
    );
28

```

Adjust nPhi and nTheta based on the required angular resolution. Higher values give more accurate directional resolution but increase computational cost significantly (total rays = $4 \times n\text{Phi} \times n\text{Theta}$).

3.5.2 Boundary Conditions

Set up the 0/IDefault file for each ray and band. The intensity boundary conditions are:

3.5.3 Numerical Schemes

28

Configure system/fvSchemes for the RTE solve. The key scheme is the divergence scheme for the ray transport term, which must use upwind differencing for stability:

28

```
divSchemes { div(Ji,li_h) bounded Gauss upwind; }
```

28

For system/fvSolution, use GAMG or PBiCGStab solvers for the intensity equations:

29

```
solvers { "ILambda.*" { solver GAMG; smoother symGaussSeidel;
```

29

```
tolerance 1e-6; relTol 0.01; } }
```

29

3.5.4 Running the Solver

29

For a serial run:

29

```
myRadiationSolver
```

29

For parallel execution:

29

```
decomposePar
```

29

```
mpirun -np 4 myRadiationSolver -parallel
```

29

```
reconstructPar
```

29

Monitor convergence by watching the residuals for each ILambda field. The RTE is essentially a steady-state problem — iterate until residuals drop below the specified tolerance.

29

3.6 Phase 6: Post-Processing and IR Signature Extraction

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3.6.1 Extracting Intensity at the Sensor

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After the solver converges, the intensity fields (ILambda_bandX_rayY) are written for each band and direction. To extract the total IR intensity arriving at the sensor patch:

30

Method 1 — postProcess utility: Use OpenFOAM's built-in patch integration:

30

```
postProcess -func 'patchIntegrate(patch=sensor,
field=ILambda_band0_ray3)'
```

30

Method 2 — Custom function object: Write a function object that sums over all rays directed toward the sensor patch and integrates over all bands to compute the total received IR power (in Watts) and the spectral distribution across bands.

30

3.6.2 Visualization in ParaView

30

Open the case in ParaView to visualize:

30

3.6.3 Interpreting Results

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The output gives radiation intensity in $\text{W/m}^2/\text{sr}$ for each band at each point on the sensor. Integrate over the sensor area to get total received power. Compare the spectral distribution across bands to identify which species (CO_2 vs H_2O) and which band (MWIR vs LWIR) contribute most to the IR signature. This information directly informs IR suppression system design.

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4. Complete File Structure Reference

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The following is the complete directory tree for the OpenFOAM case:

31

irSignature/

31

├── 0/

31

| └── T # Temperature field

31

| └── p # Pressure field

31

| └── XH2O # H2O mole fraction

31

| └── XCO2 # CO2 mole fraction

31

| └── XCO # CO mole fraction

31

| └── kappaBand0 # Absorption coeff, band 0

31

| └── kappaBand1 # Absorption coeff, band 1

31

| └── kappaBand2 # Absorption coeff, band 2

31

| └── IDefault # Default intensity BC

31

└── constant/

31

| └── polyMesh/ # Mesh files

31

```

|   └── radiationProperties # fvDOM + band config
31
|   └── system/
31
|   └── controlDict        # Solver, time, I/O
31
|   └── fvSchemes          # Discretization schemes
31
|   └── fvSolution         # Linear solvers
31
|   └── blockMeshDict      # Mesh definition
31
|   └── topoSetDict        # Region extraction
31
|   └── decomposeParDict   # Parallel decomposition
31
|   └── hitemp_data/
32
|   └── H2O.par            # Raw HITEMP data
32
|   └── CO2.par
32
|   └── kappa_lookup.dat   # Processed lookup table
32
|   └── scripts/
32
|   └── process_hitemp.py  # HITEMP processing
32

```


|—— map_plume_data.py # Data mapping utility

32

|—— extract_ir.py # Post-processing

32

5. Phase-Wise Task Checklist

33

6. Applications and Significance

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Stealth Enhancement: Understanding the IR signature enables optimization of aircraft design and development of IR suppression systems (e.g., modified nozzle geometries, coolant injection) to reduce detectability by MANPADS and IR-guided missiles.

35

Rocket Design Optimization: The solver can evaluate different exhaust nozzle designs and propellant compositions to determine which configurations minimize the IR signature while maintaining performance.

35

Real-Time Monitoring: Integration with sensor systems enables continuous monitoring of plume signatures during flight testing or operational missions.

35

Industrial Emission Analysis: The non-gray RTE solver can be applied beyond aerospace to study IR signatures from industrial stacks, power plant exhausts, and other sources of non-gray gaseous emission.

35

Computational Modeling Advancement: The modified fvDOM solver contributes a reusable, open-source tool for the CFD community, enabling non-gray radiation analysis within the widely-used OpenFOAM framework.

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7. References

35

1. Project Overview

1.1 Problem Statement

Infrared (IR) guided Anti-Aircraft Missiles, particularly Man Portable Air Defense Systems (MANPADS), pose a significant threat to military aircraft. The exhaust plume ejected from aircraft engines is one of the primary sources of IR signature, making the aircraft detectable by adversary sensors. This project aims to develop a computational feature within the OpenFOAM framework to estimate the IR radiation emitted by the exhaust plume of a military aircraft.

1.2 Objective

The primary objective is to build a standalone IR signature estimation solver in OpenFOAM that can compute the spectral radiation intensity arriving at a ground-based sensor from an aircraft exhaust plume. The solver must handle non-gray (wavelength-dependent) radiation behavior and utilize spectroscopic data from the HITEMP database for participating gaseous species (CO_2 , H_2O , CO).

1.3 Scope of Work

- Generate computational meshes for plume domain and atmosphere domain extending to a ground sensor
- Map plume flow-field data (temperature, pressure, species mole fractions) onto the computational domain
- Process HITEMP spectroscopic database to compute band-averaged absorption coefficients
- Modify OpenFOAM's gray fvDOM radiation solver to support non-gray (multi-band) spectral radiation
- Solve the Radiative Transfer Equation (RTE) across multiple IR bands and extract signature at the sensor
- Post-process results to compute total received IR power and spectral distribution at the sensor location

1.4 Software and Tools Required

Tool	Version	Purpose
OpenFOAM	v2112 or later	CFD framework and RTE solver
ParaView	5.x	Visualization and post-processing
Python 3	3.8+	HITEMP data processing scripts
C++ Compiler	GCC 9+	Custom solver compilation
HITEMP Database	2010/2020	Spectroscopic absorption data
blockMesh / snappyHexMesh	Built-in	Mesh generation

Git	Latest	Version control
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2. Theoretical Background

2.1 Radiative Transfer Equation (RTE)

The radiative transfer equation for a participating medium describes how spectral intensity changes along a ray path due to absorption, emission, and scattering. The general monochromatic RTE is:

$$dI_{\eta}/ds = \kappa_{\eta} \cdot I_{b\eta} - \beta_{\eta} \cdot I_{\eta} + (\sigma_{\eta}/4\pi) \int I_{\eta}(\hat{s}_i) \Phi_{\eta}(\hat{s}_i, \hat{s}) d\Omega_i$$

Where I_{η} is the spectral intensity along direction $\hat{s}$, $I_{b\eta}$ is the Planck blackbody function, κ_{η} is the spectral absorption coefficient, σ_{η} is the spectral scattering coefficient, β_{η} is the extinction coefficient ($\kappa_{\eta} + \sigma_{\eta}$), and Φ_{η} is the scattering phase function.

For gaseous media in aircraft plumes, scattering is negligible compared to absorption and emission. The simplified RTE becomes:

$$dI_{\eta}/ds = \kappa_{\eta} \cdot I_{b\eta} - \kappa_{\eta} \cdot I_{\eta}$$

This equation states that intensity increases by emission ($\kappa_{\eta} \cdot I_{b\eta}$) and decreases by absorption ($\kappa_{\eta} \cdot I_{\eta}$) along the path.

2.2 Gray vs. Non-Gray Radiation

Gray Radiation: Assumes the absorption coefficient κ is constant across all wavelengths. The emitted radiation follows a scaled version of the Stefan-Boltzmann law. This simplification allows a single RTE solve but is inaccurate for gaseous media where absorption is strongly wavelength-dependent.

Non-Gray Radiation: Absorption and emission properties vary with wavelength. Gaseous species like CO_2 and H_2O absorb and emit radiation in specific spectral bands, not uniformly across all wavelengths. Accurate IR signature estimation requires solving the RTE separately for each spectral band of interest, using wavelength-dependent absorption coefficients from spectroscopic databases.

2.3 Important IR Bands

Band	Wavelength	Significance
SWIR	1.4 – 3.0 μm	Sensitive to water vapor; assesses H_2O content in the plume
MWIR	3.0 – 5.0 μm	Temperature distribution; CO_2 and SO_2 detection; primary band for IR signature
LWIR	8.0 – 14.0 μm	Soot particle emissions; environmental impact assessment

2.4 HITEMP Database

The High-Temperature Molecular Spectroscopic Database (HITEMP) provides line-by-line spectroscopic data for molecular species at high temperatures (accurate up to 4000 K and 80 bar). Each spectral line transition is stored in 160-character ASCII format containing parameters such as molecule number, vacuum wavenumber, line intensity, air-broadened halfwidth, self-broadened halfwidth, lower-state energy, and temperature-dependence coefficient. The reference conditions are 296 K and 1 atm. Mathematical models are required to convert these reference parameters to actual thermodynamic conditions encountered in the plume.

3. Complete Implementation Steps

This section provides a detailed, step-by-step guide for implementing the IR signature estimation feature in OpenFOAM, from initial setup to final post-processing.

3.1 Phase 1: Environment Setup

3.1.1 OpenFOAM Installation

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Create the project directory structure following OpenFOAM conventions:

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The irSignature directory will be the main case directory. The hitemp_data folder will hold processed absorption coefficient tables, and scripts will contain Python utilities for HITEMP processing and data mapping.

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Key file: `system/blockMeshDict`

Define vertices for a cylinder aligned along the x-axis. Typical dimensions: length 0.3–0.5 m (depending on plume data extent), radius 0.015 m (based on the mole fraction profiles showing where species concentrations drop to near-zero). Use grading to concentrate cells near the axis where gradients are steepest.

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blockMesh
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After running `blockMesh`, verify the mesh quality:

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checkMesh
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Look for no failed checks, reasonable cell volumes, and acceptable non-orthogonality values.

3.2.2 Extracting Region of Interest

The full plume mesh may contain regions where species concentrations are negligible (contributing zero IR emission). Extract only the active region using `topoSet` and `subsetMesh`.

Step 1: Examine the mole fraction profiles from the plume CFD data. Identify the axial and radial extent where H_2O and CO_2 mole fractions are above a threshold (e.g., 1% of peak).

Step 2: Create `system/topoSetDict` defining a `cylinderToCell` source with the identified dimensions.

Step 3: Run the extraction:

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topoSet
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This produces a trimmed mesh containing only the radiatively active plume region.

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The atmosphere mesh extends from the plume region boundary down to the ground-based sensor. This domain captures the attenuation of the plume's IR radiation as it propagates through the atmosphere.

Method: Create a separate OpenFOAM case with `blockMesh` defining a conical or cylindrical volume from the plume exit to the sensor location (typically 1–5 km away). The mesh can be coarser than the plume mesh since atmospheric properties vary slowly.

Set atmospheric conditions in this region: ambient temperature (~288–300 K depending on altitude), atmospheric pressure (1 atm at ground, decreasing with altitude), and ambient H₂O mole fraction (~1–2% near ground).

3.2.4 Merging the Meshes

Combine the plume mesh and atmosphere mesh into a single computational domain using OpenFOAM's mesh manipulation utilities:

```
mergeMeshes . ../atmosphereCase -overwrite  
  
stitchMesh -overwrite masterPatch slavePatch
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After merging, run checkMesh again to verify the combined mesh is valid. Pay attention to the interface between plume and atmosphere regions — ensure faces are properly connected with no gaps or overlaps.

3.3 Phase 3: Data Mapping

3.3.1 Mapping Plume Flow-Field Data

The plume flow-field data (temperature T , pressure p , and species mole fractions X_{H_2O} , X_{CO_2} , X_{CO}) from a prior CFD simulation or experimental measurements must be mapped onto the computational mesh.

Option A — mapFields: If the source data is from another OpenFOAM case, use the built-in mapFields utility:

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// For each cell: T[cellI] = interpolate(x, r, tempTable)

// Similarly for p, XH2O, XCO2, XCO
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After mapping, the 0/ directory should contain the following volScalarField files: T , p , X_{H_2O} , X_{CO_2} , X_{CO} . Verify the mapping by opening the case in ParaView and visually inspecting the fields.

3.3.2 Processing HITEMP Data

This is one of the most computationally intensive and critical steps. The goal is to convert the line-by-line HITEMP data into band-averaged absorption coefficients that can be used by the RTE solver.

Step 1 — Parse the .par files: Read the 160-character fixed-width ASCII format. Extract for each spectral line: wavenumber (ν), line intensity (S) at 296 K, lower-state energy (E''), air-broadened halfwidth (γ_{air}), self-broadened halfwidth (γ_{self}), and temperature-dependence exponent (n).

Step 2 — Temperature scaling: Scale the line intensity from the reference temperature (296 K) to the local temperature T using the partition function ratio and the Boltzmann factor:

$$S(T) = S(296) * [Q(296)/Q(T)] * \exp(-c_2 * E''/T) / \exp(-c_2 * E''/296) \\ * [1 - \exp(-c_2 * \nu/T)] / [1 - \exp(-c_2 * \nu/296)]$$

Step 3 — Line broadening: Calculate the Voigt profile for each line, accounting for pressure broadening (γ_{air} , γ_{self}) and Doppler broadening at the local temperature.

Step 4 — Band averaging: Sum contributions from all spectral lines within each band of interest (e.g., 2000–3333 cm^{-1} for the MWIR 3–5 μm band, 1000–2000 cm^{-1} for the strongest fundamental band) to get a band-averaged absorption coefficient.

Step 5 — Generate lookup table: Create a lookup table of κ as a function of (T , p , X_{species}). Store this as a structured text file that the OpenFOAM solver can read at runtime.

3.3.3 Mapping Absorption Coefficients onto the Mesh

Using the lookup table from the previous step and the local T , p , and mole fraction values at each cell center, compute the absorption coefficient κ for each spectral band at every cell. Write these as volScalarField files (e.g., kappaBand0, kappaBand1, etc.) in the 0/ directory. You can use a custom utility or a codedFixedValue approach for this mapping.

3.4 Phase 4: Modifying the Radiation Solver

3.4.1 Isolating the fvDOM Solver

OpenFOAM's built-in fvDOM (finite volume Discrete Ordinates Method) solves the RTE in gray mode by default. The first step is to copy it into your user directory for modification:

```
mkdir -p $WM_PROJECT_USER_DIR/src/myRadiation
cp -r $FOAM_SRC/thermophysicalModels/radiation/
    radiationModels/fvDOM/* \
    $WM_PROJECT_USER_DIR/src/myRadiation/
```

Copy the associated Make/ files and adjust include paths to compile against the OpenFOAM libraries.

3.4.2 Adding Multi-Band (Non-Gray) Support

The core modification is to loop over multiple spectral bands instead of solving a single gray RTE. In the main solve routine of fvDOM.C, implement the following changes:

Band definition: Read band limits (wavenumber ranges) from constant/radiationProperties. Store as a list of pairs: bandLow[i], bandHigh[i].

Planck function per band: For each band, compute the Planck function integrated over the band limits at each cell's temperature. This gives the local blackbody emission for that specific spectral band.

Band-specific κ : Read or compute the band-averaged absorption coefficient for each band from the mapped fields or lookup table.

RTE solve loop: For each band b and each discrete ordinate direction d , solve:

$$\text{div}(\mathbf{J}_d, I_{\{b,d\}}) + \kappa_b * I_{\{b,d\}} = \kappa_b * I_{b_band}(T)$$

Where $\mathbf{J}_d$ is the direction vector for ordinate d , $I_{\{b,d\}}$ is the intensity for band b in direction d , κ_b is the band-averaged absorption coefficient, and $I_{b_band}(T)$ is the Planck function integrated over band b .

Storage: Create volScalarField arrays indexed by [band][ray] to store the intensity fields for each band and direction.

3.4.3 Compilation

Create a Make/files and Make/options in your modified radiation directory. Compile the custom library:

```
cd $WM_PROJECT_USER_DIR/src/myRadiation
wmake libso
```

This produces a shared library (e.g., libmyRadiation.so) that you reference in your case's system/controlDict via the libs directive.

3.5 Phase 5: Case Setup and Simulation

3.5.1 constant/radiationProperties

Configure the radiation model to use your modified fvDOM with band definitions:

```
radiation          on;

radiationModel     fvDOM;

fvDOMCoeffs
{
    nPhi    4;          // azimuthal angles
    nTheta  4;          // polar angles
    maxIter 10;
    tolerance 1e-4;
}

nBands  3;

bandLimits
(
    (2000 3333)    // MWIR: 3-5 um
    (1000 2000)    // Strongest fundamental band
    (714  1250)    // LWIR: 8-14 um
);
```

Adjust nPhi and nTheta based on the required angular resolution. Higher values give more accurate directional resolution but increase computational cost significantly (total rays = $4 \times n\text{Phi} \times n\text{Theta}$).

3.5.2 Boundary Conditions

Set up the 0/IDefault file for each ray and band. The intensity boundary conditions are:

- Inlet (plume entry): Calculated from local blackbody emission, or set to zero if no upstream radiation source
- Outlet (far field): Zero gradient or fixed value at ambient blackbody intensity
- Sensor patch: Calculated — this is where you extract the arriving intensity
- Outer boundaries: Zero incoming intensity (cold surroundings) or ambient blackbody temperature

3.5.3 Numerical Schemes

Configure system/fvSchemes for the RTE solve. The key scheme is the divergence scheme for the ray transport term, which must use upwind differencing for stability:

```
divSchemes { div(Ji,Ii_h) bounded Gauss upwind; }
```

For system/fvSolution, use GAMG or PBiCGStab solvers for the intensity equations:

```
solvers { "ILambda.*" { solver GAMG; smoother symGaussSeidel;  
    tolerance 1e-6; relTol 0.01; } }
```

3.5.4 Running the Solver

For a serial run:

```
myRadiationSolver
```

For parallel execution:

```
decomposePar
```

```
mpirun -np 4 myRadiationSolver -parallel
```

```
reconstructPar
```

Monitor convergence by watching the residuals for each ILambda field. The RTE is essentially a steady-state problem — iterate until residuals drop below the specified tolerance.

3.6 Phase 6: Post-Processing and IR Signature Extraction

3.6.1 Extracting Intensity at the Sensor

After the solver converges, the intensity fields (ILambda_bandX_rayY) are written for each band and direction. To extract the total IR intensity arriving at the sensor patch:

Method 1 — postProcess utility: Use OpenFOAM's built-in patch integration:

```
postProcess -func 'patchIntegrate(patch=sensor, field=ILambda_band0_ray3)'
```

Method 2 — Custom function object: Write a function object that sums over all rays directed toward the sensor patch and integrates over all bands to compute the total received IR power (in Watts) and the spectral distribution across bands.

3.6.2 Visualization in ParaView

Open the case in ParaView to visualize:

- Temperature and species concentration fields to verify data mapping quality
- Absorption coefficient fields for each band to confirm HITEMP data processing
- Intensity fields along ray directions to see the radiation propagation pattern
- Intensity distribution on the sensor patch to determine the spatial IR signature footprint

```
paraFoam
```

3.6.3 Interpreting Results

The output gives radiation intensity in $\text{W/m}^2/\text{sr}$ for each band at each point on the sensor. Integrate over the sensor area to get total received power. Compare the spectral distribution across bands to identify which species (CO_2 vs H_2O) and which band (MWIR vs LWIR) contribute most to the IR signature. This information directly informs IR suppression system design.

4. Complete File Structure Reference

The following is the complete directory tree for the OpenFOAM case:

```

irSignature/

├── 0/

|   ├── T                # Temperature field
|   ├── p                # Pressure field
|   ├── xH2O             # H2O mole fraction
|   ├── xCO2             # CO2 mole fraction
|   ├── xCO              # CO mole fraction
|   ├── kappaBand0       # Absorption coeff, band 0
|   ├── kappaBand1       # Absorption coeff, band 1
|   ├── kappaBand2       # Absorption coeff, band 2
|   └── IDefault         # Default intensity BC

├── constant/

|   ├── polyMesh/        # Mesh files
|   └── radiationProperties # fvDOM + band config

├── system/

|   ├── controlDict      # Solver, time, I/O
|   ├── fvSchemes        # Discretization schemes
|   ├── fvSolution       # Linear solvers
|   ├── blockMeshDict    # Mesh definition
|   ├── topoSetDict      # Region extraction
|   └── decomposeParDict # Parallel decomposition

```

```
|— hitemp_data/
|   |— H2O.par           # Raw HITEMP data
|   |— CO2.par
|   └— kappa_lookup.dat  # Processed lookup table
└— scripts/
    |— process_hitemp.py  # HITEMP processing
    |— map_plume_data.py  # Data mapping utility
    └— extract_ir.py      # Post-processing
```


5. Phase-Wise Task Checklist

#	Task	Status	Owner
PHASE 1: ENVIRONMENT SETUP			
1	Install OpenFOAM v2112+	Pending	Shubham
2	Set up project directory structure	Pending	Shubham
3	Download HITEMP .par files (H ₂ O, CO ₂ , CO)	Pending	Shubham
PHASE 2: MESH GENERATION			
4	Create initial plume mesh (blockMesh)	Pending	Shubham
5	Extract region of interest (topoSet + subsetMesh)	Pending	Shubham
6	Create atmosphere mesh	Pending	Shubham
7	Merge plume + atmosphere meshes	Pending	Shubham
8	Verify merged mesh quality (checkMesh)	Pending	Shubham
PHASE 3: DATA MAPPING			
9	Map T, p, species fields onto plume mesh	Pending	Shubham
10	Process HITEMP data into absorption coefficients	Pending	Shubham
11	Generate κ lookup table (T, p, X)	Pending	Shubham
12	Map κ for each band onto mesh	Pending	Shubham
PHASE 4: SOLVER MODIFICATION			
13	Copy fvDOM to user directory	Pending	Shubham
14	Implement multi-band loop	Pending	Shubham
15	Add Planck band integration	Pending	Shubham
16	Compile modified library (wmake)	Pending	Shubham
PHASE 5: SIMULATION			
17	Configure radiationProperties	Pending	Shubham
18	Set boundary conditions for all bands	Pending	Shubham
19	Run gray validation case first	Pending	Shubham
20	Run full non-gray multi-band simulation	Pending	Shubham
PHASE 6: POST-PROCESSING			
21	Extract intensity at sensor patch	Pending	Shubham
22	Compute total IR power per band	Pending	Shubham

23	Visualize results in ParaView	Pending	Shubham
24	Compare MWIR vs LWIR contributions	Pending	Shubham
25	Document results and conclusions	Pending	Shubham

6. Applications and Significance

Stealth Enhancement: Understanding the IR signature enables optimization of aircraft design and development of IR suppression systems (e.g., modified nozzle geometries, coolant injection) to reduce detectability by MANPADS and IR-guided missiles.

Rocket Design Optimization: The solver can evaluate different exhaust nozzle designs and propellant compositions to determine which configurations minimize the IR signature while maintaining performance.

Real-Time Monitoring: Integration with sensor systems enables continuous monitoring of plume signatures during flight testing or operational missions.

Industrial Emission Analysis: The non-gray RTE solver can be applied beyond aerospace to study IR signatures from industrial stacks, power plant exhausts, and other sources of non-gray gaseous emission.

Computational Modeling Advancement: The modified fvDOM solver contributes a reusable, open-source tool for the CFD community, enabling non-gray radiation analysis within the widely-used OpenFOAM framework.

7. References

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